



ScienceDirect®

Journal of Functional Foods

Volume 134, November 2025, 107030

Prebiotic activity of functional whole mushroom powders compared to inulin in SHIME® model colonic simulations

Julie Daoust ^a  , Jay Schmalz ^a , Chloë Rotsaert ^b , Chelsea Vannieuwenhuyse ^b ,
Jelle De Medts ^b , Lieven Van Meulebroek ^{b c} , Cindy Duysburgh ^b 

[Show more](#) 

 Outline |  Share  Cite

<https://doi.org/10.1016/j.jff.2025.107030> 

[Get rights and content](#) 

Under a Creative Commons [license](#) 

Open access

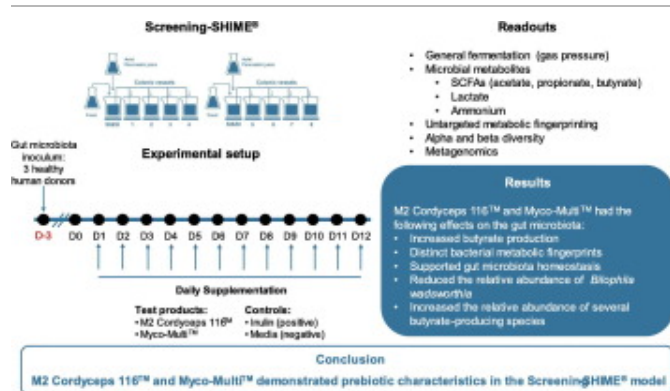
Highlights

- M2 Cordyceps 116™ and Myco-Multi™ supplementation increased butyrate production.
- Distinct bacterial metabolic fingerprints were observed with each test product.
- The test products decreased the relative abundance of *Bilophila wadsworthia*.
- The test products increased the relative abundance of butyrate-producing species.

Abstract

Minimally processed, whole mushroom powders (M2 Cordyceps 116™ and Myco-Multi™) were evaluated for their effects on the human gut microbiota using the Screening-Simulator of the Human Intestinal Microbial Ecosystem (SHIME®) in vitro model. A negative control (blank media) and positive control (inulin) were included. The test products were administered daily for 12 days. Both M2 Cordyceps 116™ and Myco-Multi™ enhanced acetate (significant for both test products) and butyrate (significant for M2 Cordyceps 116™) production versus the negative and positive controls. Distinct bacterial metabolic fingerprints were observed for the two test products and controls. The effect on microbial species richness and evenness was mild, indicating that the test products supported gut microbiome homeostasis. Beta diversity analysis showed distinct clustering with each test product and control, probably attributable to the stimulation of several butyrate-producing bacterial species following product supplementation. M2 Cordyceps 116™ and Myco-Multi™ demonstrated distinct prebiotic characteristics in the Screening-SHIME® model.

Graphical abstract



[Download: Download high-res image \(239KB\)](#)

[Download: Download full-size image](#)

[<](#) Previous

Next [>](#)

Keywords

Cordyceps; Functional mushroom; Metagenomics; Mycelium; Prebiotic; Butyrate

Abbreviations

CC, M2 Cordyceps 116™; CV-ANOVA, cross-validation analysis of variance; DAPC, Discriminant Analysis of Principal Components; FC, fold change; GLP-1, glucagon-like peptide-1; LA-REIMS, Laser-Assisted Rapid Evaporative Ionization Mass Spectrometry; LDA, Linear Discriminant Analysis; LEfSe, Linear Discriminant Effect Size; MM, Myco-Multi™; OPLS-DA, Orthogonal Partial Least

Squares Discriminant Analysis; PBS, phosphate buffered saline; PC, positive control; PCA-X, Unsupervised Principal Component Analysis; PYY, peptide YY; SCFA, short-chain fatty acid; SHIME®, Simulator of the Human Intestinal Microbial Ecosystem; UHPLC-HRMS, ultra-high performance liquid chromatography-high-resolution mass spectrometry system

1. Introduction

Prebiotics are defined as “compounds or ingredients which are utilized by the gut microbiota, producing a health or performance benefit that arises from alteration of the composition and/or activity of the microbiome” (Deehan et al., 2024). These typically consist of non-digestible oligosaccharides, which resist digestion and absorption while transiting the small intestine but are completely or partially fermented by bacteria via saccharolytic fermentation in the large intestine/colon (Davani-Davari et al., 2019). Microbial metabolites are produced as byproducts of both saccharolytic and proteolytic fermentation; a major metabolic byproduct of saccharolytic fermentation is short-chain fatty acids (SCFAs), which are involved in several aspects of human health (Davani-Davari et al., 2019; Roberfroid, 1997). Prebiotics can also affect the composition of the gut microbiome by supporting the growth of saccharolytic bacteria, such as *Bifidobacterium* (Carlson et al., 2018; Liu et al., 2016). A shift to a greater abundance of saccharolytic bacteria within the gut microbiome leads to increased production of SCFAs such as acetate, propionate, and butyrate. Inulin, a well-established prebiotic, selectively modulates the composition and activity of the colonic microbiota. This has beneficial effects, including improvement of intestinal barrier function, laxation, and insulin sensitivity (Hughes et al., 2022; Vandeputte et al., 2017).

Mushrooms have been used in traditional medicine practices to address a wide variety of health conditions. Early records of the Materia Medica demonstrate the use of mushrooms for treating disease. The prebiotic effect and modulation of the gut microbiome by edible and medicinal mushrooms has become the subject of extensive research in recent years (Araujo-Rodrigues et al., 2024). Mushroom-derived bioactive compounds have shown beneficial prebiotic effects in studies using in vitro fermentation models, animal models, and in humans (Yu et al., 2023; Zhao et al., 2023). Many of the health benefits that medicinal mushrooms are known for have demonstrated microbiota-mediated mechanisms. For example, mouse studies demonstrated that the sleep health benefit of Reishi mushroom was mediated via the gut microbiome (Yao et al., 2021). The cognitive benefit of Lion's Mane may also be related to its effect on SCFA-producing intestinal bacteria (Zhuang et al., 2023). *Cordyceps militaris* has demonstrated gut microbiome modulation in animal studies for hyperlipidemia, diabetes, and chronic obstructive pulmonary disease (Gamage et al., 2018; W. Zhang, et al., 2025; Zheng et al., 2022). The immunomodulatory effect, hypolipidemic effect, and hypoglycemic effect of several mushroom species have been linked to their influence on the microbiome (Bai et al., 2024; Diling et al., 2017). Novel functional products that include prebiotic mushrooms have been developed and tested (Atik et al., 2024; Cortes Bordoy et al., 2023;

[Moumita, et al., 2022](#)).

To date, research has focused on extracted fractions or specific mushroom parts – fruit body, spore, or mycelium. Research evaluating the effect of whole, full spectrum mushrooms or blends/combinations of mushroom species on the microbiome is limited. It has been shown that mushroom fruit body powders have positive influence on the gut microbiota and production of SCFAs ([Mitsou et al., 2020](#)). Fresh fruit bodies from *Agaricus bisporus* were reported to modulate the gut microbiota compared with a diet of protein-matched quantities of meat in a crossover study ([Hess et al., 2018](#)). A blend of three mushroom species was studied via an in vitro colon model and demonstrated a dose-dependent increase in butyrate production by the microbiome ([Verhoeven et al., 2021](#)). This research investigates the effect of minimally processed full-spectrum mushroom powders and blends containing multiple mushroom species.

Using whole mushroom powders as prebiotic ingredients can provide additional functional benefits over isolated polysaccharides fractions like inulin, that align with rising consumer demand for functional, minimally processed whole food ingredients. From an economic standpoint, this trend for whole-food ingredients ties into the surging functional-mushroom market. The global medicinal mushroom market is projected to climb from USD 34.22 billion in 2024 to USD 59.41 billion by 2031, growing at a compound annual rate of **8.2 %** ([Coherent Market Insights, 2025](#)). This rapid growth reflects consumer appetite for multifunctional natural products—driven by the potential benefits of mushrooms not only to support digestive health but also for their immunomodulating, anti-inflammatory, antioxidants, and even neuroprotective effects ([Liu et al., 2023](#); [Singh et al., 2025](#)). Hence, while inulin remains a well-established, cost-effective prebiotic ingredient, whole mushrooms can offer additional benefits by combining prebiotic effects with other functional benefits making them increasingly attractive in functional foods products.

In vitro models of the gastrointestinal system allow researchers to perform mechanistic studies in a well-controlled environment, limiting the influence of external factors. Additionally, multiple conditions can be tested simultaneously. The standard Simulator of the Human Intestinal Microbial Ecosystem (SHIME®) model allows for the in vitro evaluation of the effects of repeated test product dosing on the gut microbiome over the course of 7 weeks (2-week stabilization period, 2-week control period, 3-week treatment period), and has been validated with in vivo human data on both human microbial function and composition ([Molly et al., 1993](#); [Possemiers et al., 2004](#)). The Screening-SHIME® is a newly developed iteration of the standard SHIME® that also allows for the evaluation of the effect of repeated test product administration on the gut microbiome, but with a faster turnaround time of 16 days or less (1-day inoculation period, 3-day stabilization period, 8-to-12-day treatment period). In this model, the test product and untreated control are run in parallel.

This study aimed to determine the prebiotic effects of repeated administration of functional whole mushroom powders specifically *Cordyceps militaris* (M2 Cordyceps 116™), and a blend of 10 species of functional mushrooms (Myco-Multi™) and to evaluate how they compare to the well-studied

prebiotic, inulin, using the Screening-SHIME® model. Bacterial fermentation, microbial metabolite production, and changes in the microbial community composition after repeated administration of M2 Cordyceps 116™ and Myco-Multi™, or inulin (positive control) were evaluated over a 12-day treatment period.

2. Material and methods

2.1. Test products

M2 Ingredients Inc. (Vista, CA, USA) provided the functional whole mushroom test materials, M2 Cordyceps 116™ and Myco-Multi™. All mushroom species were individually cultured on whole certified organic oats (*Avena sativa*), dehydrated gently to break down the chitinous cell walls of the fungi, and then milled into a powder. Identities were confirmed with species positive identification specifications using DNA sequencing of master tissue culture, taxonomic and visual monitoring of morphology, and growth metrics during the growing cycle.

The M2 Cordyceps 116™ test material is the patented strain *Cordyceps militaris* M2–116–04 and contains mycelium, stroma, and fruit body. The Myco-Multi™ blend consists of ten single species mushroom powders blended together: Chaga M2–100–09 (*Inonotus obliquus*) mycelium; King Trumpet M2–102–08 (*Pleurotus eryngii*) mycelium and fruit body; *Cordyceps militaris* M2–116–04 mycelium, stroma, and fruit body; Maitake M2–100–07 (*Grifola frondosa*) mycelium; Shiitake M2–104–11 (*Lentinula edodes*) mycelium and primordia; Lion's Mane M2–102–10 (*Hericium erinaceus*) mycelium and fruit body; Reishi M2–102–02 (*Ganoderma lucidum* s.l.) mycelium and fruit body; Turkey Tail M2–101–03 (*Trametes versicolor*) mycelium and primordia; *Antrodia camphorata* M2–100–05 mycelium; and *Agaricus blazei* M2–101–01 mycelium. Inulin (Frutafit®; Sensus, Roosendaal, the Netherlands) was supplemented as a positive control. As the functional whole mushroom test materials would be intended to replace prebiotic intake in vivo with regard to gastrointestinal maintenance, inulin was selected as the comparator as it represents a gold-standard prebiotic.

2.2. Fecal samples

Stool samples were collected from three healthy adult donors between the ages of 28 and 35 years (donor A: F, 28y; donor B: F, 30y; donor C: M; 35y). All fecal donors had no history of antibiotic use during the 3 months prior to their donation and had no known history of chronic gut inflammation. Upon collection, phosphate buffered saline (PBS) was added to the fecal samples in an anaerobic environment, then homogenized to create a slurry. Large particles were removed following brief centrifugation and an equal volume of optimized in-house cryoprotectant (modified from Hoefman et al. (Hoefman et al., 2013)) was added. The fecal suspensions were flash frozen in liquid nitrogen and stored at –80 °C. To initiate the experiment, fecal samples were defrosted and immediately added to the reactors. Fecal materials were collected and used as approved by the Ethics Committee of the University Hospital Ghent (reference number ONZ-2022-0267; approved on 29 July 2022).

2.3. Screening-SHIME®

This study employed the Screening-SHIME® model, which uses inoculum preparation, retention time, pH, temperature, and reactor feed composition parameters adapted from the standard SHIME® model as reported in Possemiers et al. ([Possemiers et al., 2004](#)). The Screening-SHIME® set-up includes the combination of several sets of a combined stomach and small intestine compartment followed by up to four single colon reactors run in parallel, with each colon reactor simulating a different test condition, making it possible to test multiple test products and/or experimental donors in parallel. In the current study, four sets of the Screening-SHIME® model were run, including three colonic vessels per set. In this way, three different test products (inulin, M2 Cordyceps 116™, and Myco-Multi™) were investigated versus a negative control (= fourth condition) for three different donors, resulting in a repeated-dosing experiment with twelve unique test conditions. Each donor was tested in single replicate, resulting in three biological replicates for each test condition. Following fecal sample inoculation, the colon reactors were set to simulate the colon, with a pH of 6.15–6.40, retention time of 32 h, and a volume of 400 mL per reactor. Each colon reactor contained both luminal and mucosal colon environments. The mucosal environment was simulated by the addition of mucin-coated beads to the colon reactor as previously described by Van den Abbeele et al. (Van den Abbeele et al., 2013). On experimental Day –3 (D-3), colon reactors were inoculated with a 5 % (v/v) cryopreserved inoculum containing 20 % (w/v) fecal material from donor A, donor B, or donor C. The fecal microbial community was allowed to grow and colonize overnight. The reactors were fed and allowed to stabilize over experimental Days –2 (D-2) through 0 (D0). The treatment period was initiated on Day 1 (D1) and continued for 12 days, through Day 12 (D12). Throughout the experiment, the stomach/small intestine reactors were subsequently fed three times daily with standard SHIME® nutritional medium (14.6 g/L PDNM002B, ProDigest, Belgium; 70 mL; pH, 2.0) and pancreatic juice (12.5 g NaHCO₃, 6 g/L oxgall, and 0.9 g/L pancreatin; 30 mL) which entered the colon reactors 3 h after initiating the feeding cycle. During the treatment phase, the reactors were operated under nominal conditions and the nutritional medium was supplemented with the indicated test product once daily. No test product was added to the negative control arms. The mushroom test products were supplemented at a dose corresponding to an in vivo intake of 2 g/day and inulin, the positive control, was supplemented at a dose corresponding to an in vivo intake of 1 g/day, which matched the fiber content of the mushroom test products. The 2 g dose for the mushroom powders was selected to align with the safe recommended dose for *C. militaris* and *Ganoderma lucidum* s.l. recommended by the manufacturer following toxicology studies ([Chrysostomou, P. et al. 2024](#)).

2.4. Analysis

2.4.1. Microbial metabolic activity

Gas production was assessed using short-term colonic simulations. The short-term simulations were initiated using bacterial inoculum from the SHIME® experiment taken three days after the

treatment period was initiated to ensure the presence of pre-adapted microbial communities. The individual test products, bacterial inoculum (10 % v/v), and nutritional medium (PD01, ProDigest, Ghent, Belgium) were added to reactors and incubated under anaerobic conditions for 48 h at 37 °C with shaking (90 rpm). Total gas production was assessed after 0 h, 4 h, 6 h, 24 h, and 48 h incubation by measuring gas pressure in the reactors using a hand-held pressure indicator (WIKA, CPH6200, Lawrenceville, Georgia, USA) with a transmitter (WIKA, CPT6200). Capillary gas chromatography coupled with a flame ionization detector was used to measure levels of SCFA (acetate, propionate, and butyrate) production (De Weirldt et al., 2010). Prior to SCFA isolation, samples were thawed at 4 °C, briefly vortexed (400 rpm, 20 °C, 1 min), and 250 µL was transferred into a 12 mL test tube. SCFA isolation was achieved by liquid-liquid extraction using acetonitrile as extract solvent, with a linear dynamic range of 0.05–50 mM (limit of detection, 0.02 mM). Prior to lactate and ammonium quantification, samples were thawed at 4 °C, centrifuged (5 min, 9000 rpm), and 1000 µL supernatant was collected. Lactate concentrations were quantified using the Enzytec™ kit (R-Biopharm, Darmstadt, Germany). Colorimetric analysis using the indophenol blue spectrophotometric method as described by Duysburgh et al. was used to measure ammonium levels (Duysburgh et al., 2024). For each assay (except gas production), samples were collected on D1, D3, D5, D8, D11, and D12, in order to be able to determine acute and maximal treatment effects following repeated dosing. Measurements were performed in single repetition.

2.4.2. Untargeted metabolic fingerprinting (LA-REIMS)

Untargeted metabolic fingerprints for each sample were measured using the Laser-Assisted Rapid Evaporative Ionization Mass Spectrometry (LA-REIMS) platform (Plekhova et al., 2021). Prior to LA-REIMS analysis, samples were thawed at 4 °C, briefly vortexed (400 rpm, 20 °C, 1 min), and 100 µL was transferred into a well of a 96-well plate. For laser ablation and associated metabolite release, the LA-REIMS platform utilized a mid-infrared laser system (Opolette™ HE2940, OPOTEK, LLC, Carlsbad, California, USA). For mass analysis, a Xevo G2-XS Quadrupole Time-of-Flight mass spectrometer (Waters Corporation, Wilmslow, UK) was used. Using this platform, the 50–1200 Da *m/z*-scan range was targeted in negative ionization mode. Each measurement was done in a single repetition.

2.4.3. Metagenomics

Samples were collected (D5 and D12 from the lumen and mucosal environment) for microbial community composition assessment using shallow shotgun sequencing, which was conducted in triplicate. DNA was extracted using the DNeasy Powersoil Pro kit (Qiagen, Germantown, USA) according to manufacturer's instructions, starting from pelleted 1 mL samples for the luminal environment and 250 mg mucus for the mucosal environment. To determine the taxonomic classification of the sequences obtained through shallow shotgun sequencing, Kneaddata v0.10.0 was used for quality filtering, trimming, and host decontamination (human genome [hg37] of raw reads with the params:-.

SLIDINGWINDOW:5:22 MINLEN:100 AVGQUAL:22). Using the Genome Taxonomy Database (v214 along with Refseq genomes from fungi + protozoan + virus), which contains $\pm 100,000$ species for taxonomic classification, quality filtered reads were subjected to Kraken2 v2.1.3 (confidence threshold, 0.1) and Bracken v2.9 (read threshold, 50). Samples were filtered on a prevalence of $\geq 5\%$ and on a minimal relative abundance of $\geq 0.01\%$. Filtering was done separately for all luminal and mucosal samples.

Flow cytometry (high flow rate) was used to determine the total number of bacterial cells in each sample (BD Accuri C6 Plus Flow Cytometer; BD Biosciences, Franklin Lakes, New Jersey, USA). Bacterial cells were separated from medium debris and signal noise using a threshold level of 700 on the SYTO channel. Parent and daughter gates were set to determine all populations.

2.5. Statistical analysis

The effects of treatment on microbial metabolic activity (gas pressure, acetate, propionate, butyrate, lactate, ammonium) across donors ($n = 3$) were evaluated using per-donor measurements as replicate values and compared with the controls using paired two-sided student's *t*-tests.

Alpha-diversity was evaluated (phyloseq v1.44.0 (McMurdie, et al., 2013)) using four measures: (1) Observed taxa (species richness), (2) Chao1 (species richness), (3) Shannon diversity index (species richness and evenness), and (4) the Simpson index (species richness and evenness, with more weight given to common or dominant species). Beta-diversity analysis was conducted by hierarchically clustering Euclidean distances between samples using Ward's minimum variance method. A Discriminant Analysis of Principal Components (DAPC) plot was constructed with two discriminants and 80 % percent of retained variance in the principal components using adegenet v2.1.10 (Jombart, 2008; Jombart et al., 2010). Both analyses were run in R v4.3.1 and were plotted using ggplot2 (v3.4.2).

Linear Discriminant Effect Size (LEfSe) (Segata et al., 2011) analysis was performed on relative abundance data obtained by total-sum scaling. Linear Discriminant Analysis (LDA) scores between treatment and individual taxon abundances were calculated using MASS (v7.3.58–3). Features that met $p \leq 0.05$ for the Kruskal-Wallis and Wilcoxon tests are shown on the LEfSe tables. There were no minimal score restrictions for the LDA. A score of ≥ 2 is generally considered biologically relevant. TreeclimbR analysis was also conducted (Huang et al., 2021). Bacterial enrichments were considered statistically significant if they had a $-\log(p\text{-value}) > 1.3$. Four categories were used for taxa classification: (1) not significant and not biologically relevant ($-2 < \log_2 \text{fold change [FC]} < +2$, and $-\log_{10}[p\text{-value}] < 1.3$), (2) biologically relevant, but not statistically significant ($\log_2 \text{FC} < -2$ or $\log_2 \text{FC} > +2$, and $-\log_{10}[p\text{-value}] < 1.3$), (3) statistically significant, but not biologically relevant ($-2 < \log_2 \text{FC} < +2$, and $-\log_{10}[p\text{-value}] > 1.3$), and (4) biologically and statistically significant ($\log_2 \text{FC} < -2$ or $\log_2 \text{FC} > +2$, and $-\log_{10}[p\text{-value}] > 1.3$). LEfSe pairwise comparisons were run in R package stats4 (The R Foundation for Statistical Computing, Vienna, Austria), and LDA scores between the test product and control taxon abundances were calculated using the R package MASS

v7.3.58–3. TreeclimbR analysis was run using treeclimbR v0.1.5 and edgeR v3.42.421. Benjamini-Hochberg multiple testing correction was used, and the alpha-level was set at 0.05. Both analyses were run in R v4.3.1.

For metabolic fingerprinting (LA-REIMS) analysis, data were normalized and the multivariate statistical analysis was performed using SIMCA 17 (Sartorius, Germany). Data were pre-processed by performing log-transformation to induce normal distributions and unit variance scaling (1/standard deviation) to standardize the range of signal intensities. Unsupervised Principal Component Analysis (PCA-X) was used to assess the natural patterning of samples and identify potential outliers (based on the Hotelling's T2 criterion). Samples were differentiated by experimental conditions in a supervised fashion using Orthogonal Partial Least Squares Discriminant Analysis (OPLS-DA). Permutation testing ($n = 100$), the quality parameter $Q^2(Y)$ (≥ 0.5), and cross-validated analysis of variance (p-value < 0.05) were used to verify the validity of the OPLS-DA models (MKS Umetrics, 2012).

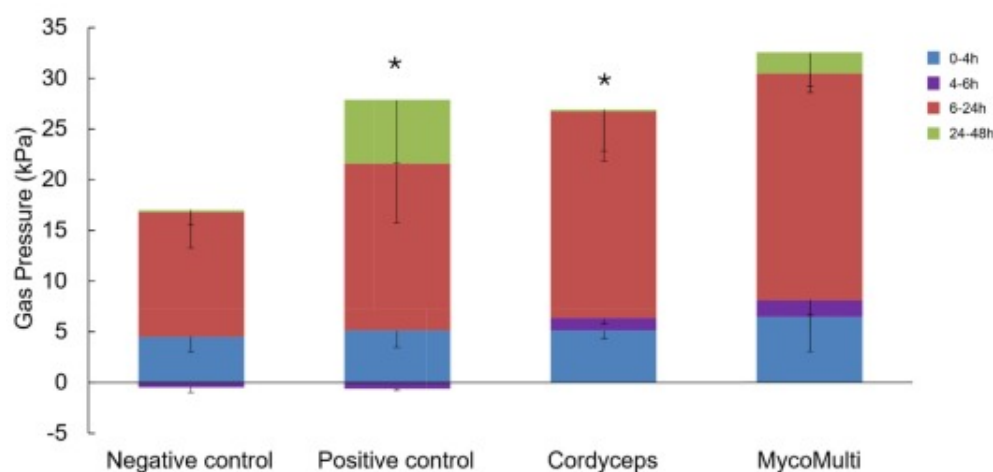
A p-value of < 0.05 was considered statistically significant.

3. Results

3.1. Metabolic activity and gas production

3.1.1. Gas production

Across donors, gas pressure increased over time in all colon reactors (Fig. 1). Compared with the negative control, gas pressure was significantly higher in the reactors treated with M2 Cordyceps 116™ and the positive control, but not Myco-Multi™ at 48 h.



[Download: Download high-res image \(56KB\)](#)

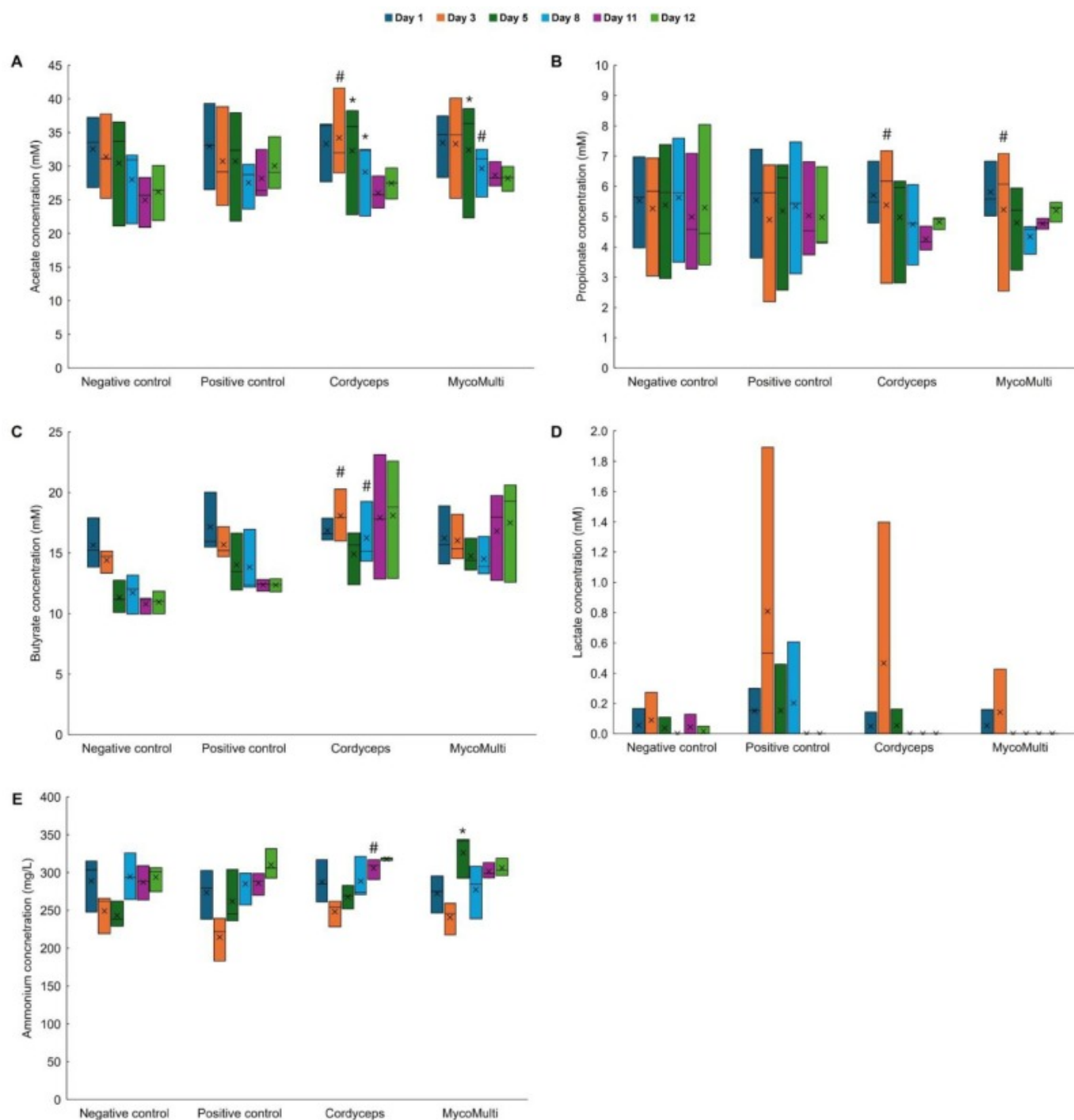
[Download: Download full-size image](#)

Fig. 1. Average gas pressure (kPa) during the time intervals of 0-4 h, 4-6 h, 6-24 h, and 24-48 h.

Microbiota of three healthy donors from the colonic vessels of the Screening-SHIME® experiment following the initiation of the treatment period were placed into short-term colonic incubations and gas pressure was measured over a 48 h period. Data are plotted as the average across donors ($n = 3$). Error bars indicate standard deviation. Paired two-tailed student's *t*-tests were used for statistical comparisons. * $p < 0.05$ versus the negative control. # $p < 0.05$ versus the positive control. SHIME®, Simulator of the Human Intestinal Microbial Ecosystem.

3.1.2. Metabolic activity

Across donors, the concentration of acetate was significantly increased on D5 with both M2 Cordyceps 116™ and Myco-Multi™ supplementation compared with the negative control (Fig. 2A). On D8, acetate levels were significantly higher with M2 Cordyceps 116™ supplementation versus the negative control. Compared with the positive control, acetate production was significantly higher in the M2 Cordyceps 116™-supplemented reactors at D3 and with Myco-Multi™ supplementation on D8. There were no significant differences in propionate production across donors between the negative control reactors and those supplemented with M2 Cordyceps 116™ or Myco-Multi™ at any timepoint (Fig. 2B). Compared with the positive control, propionate levels were significantly higher in the M2 Cordyceps 116™ and Myco-Multi™-supplemented reactors on D3. Similarly, there were no significant differences in the concentration of butyrate across donors between the negative control and the mushroom test products at any timepoint, though the levels trended higher with M2 Cordyceps 116™ and Myco-Multi™ versus the negative control (Fig. 2C). On D3 and D8, butyrate levels were significantly higher in the M2 Cordyceps 116™-supplemented reactors compared with the positive control. In the Myco-Multi™-supplemented reactors, butyrate levels were higher than the positive control on D11 and D12, but the difference did not reach significance. In general, lactate levels were stable over the study duration, except for a notable increase in the positive control and M2 Cordyceps 116™-supplemented reactors on D3, due to some variability among the donors (Fig. 2D). There were no significant differences between the negative control or positive control and test product supplementation at any timepoint. There were no differences in ammonium production across donors between the negative control and the mushroom test products at any timepoint, except for a significant increase with Myco-Multi™ supplementation versus the negative control on D5 (Fig. 2E). Similarly, limited differences were observed between the mushroom test products and the positive control, except for a significant increase with M2 Cordyceps 116™ supplementation versus the positive control on D11.



[Download: Download high-res image \(624KB\)](#)

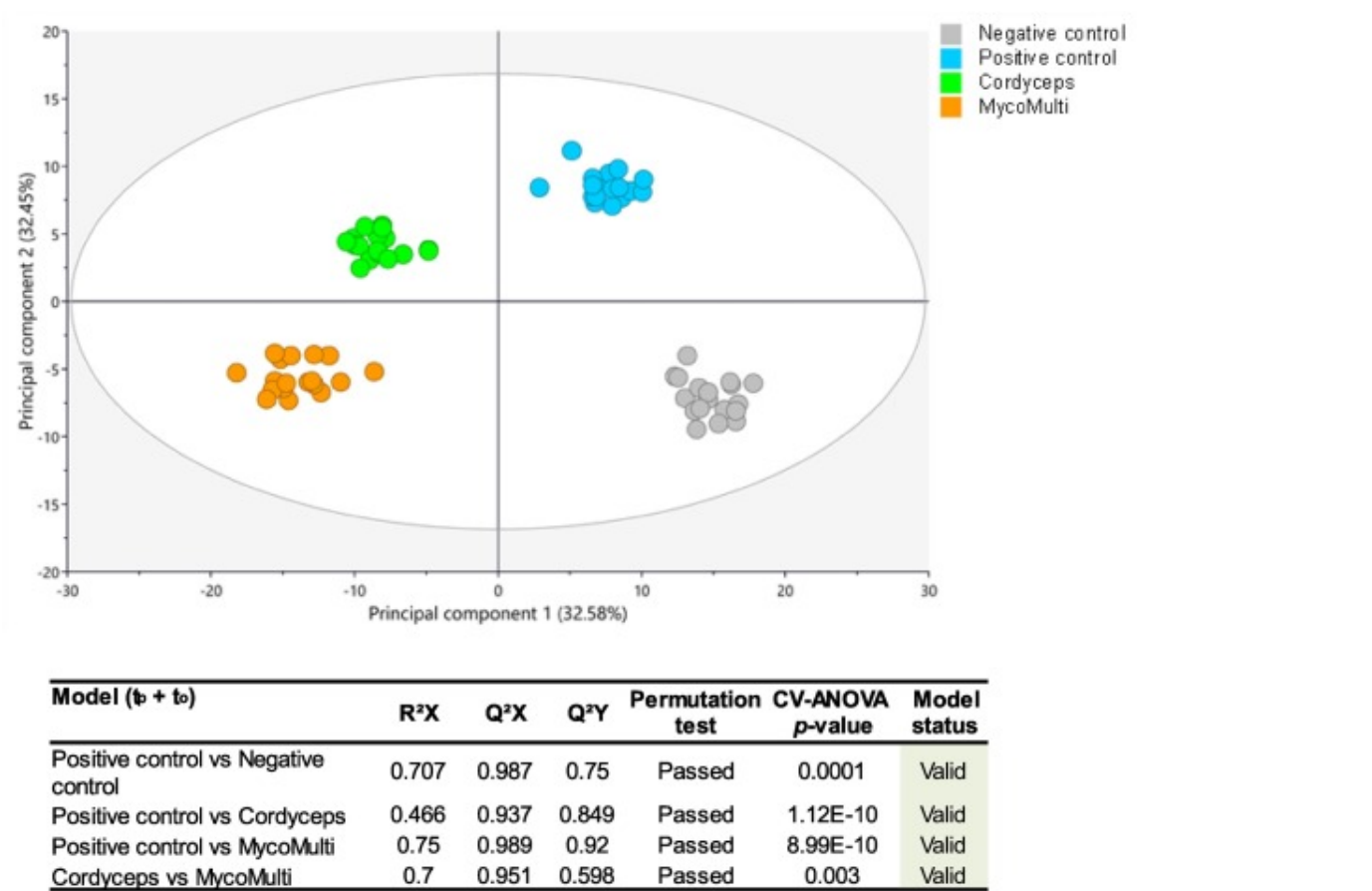
[Download: Download full-size image](#)

Fig. 2. Boxplots showing acetate (A), propionate (B), butyrate (C), lactate (D), and ammonium (E) levels at Days 1, 3, 5, 8, 11, and 12. Levels of acetate, propionate, butyrate, lactate, and ammonium in colonic fermentations collected after D1, D3, D5, D8, D11, and D12 in the Screening-SHIME® model using fecal samples from three healthy human donors. The × represents the average across donors and the horizontal line represents the median across donors (n = 3). Paired two-tailed student's t-tests were used for statistical comparisons. * $p < 0.05$ versus the negative control. # $p < 0.05$ versus the positive control.

D, day; SHIME®, Simulator of the Human Intestinal Microbial Ecosystem.

3.2. Untargeted metabolomics

The OPLS-DA score plot showed that both M2 Cordyceps 116™ and Myco-Multi™ supplementation result in different metabolic fingerprints compared with the negative and positive controls, and compared with each other (Fig. 3).



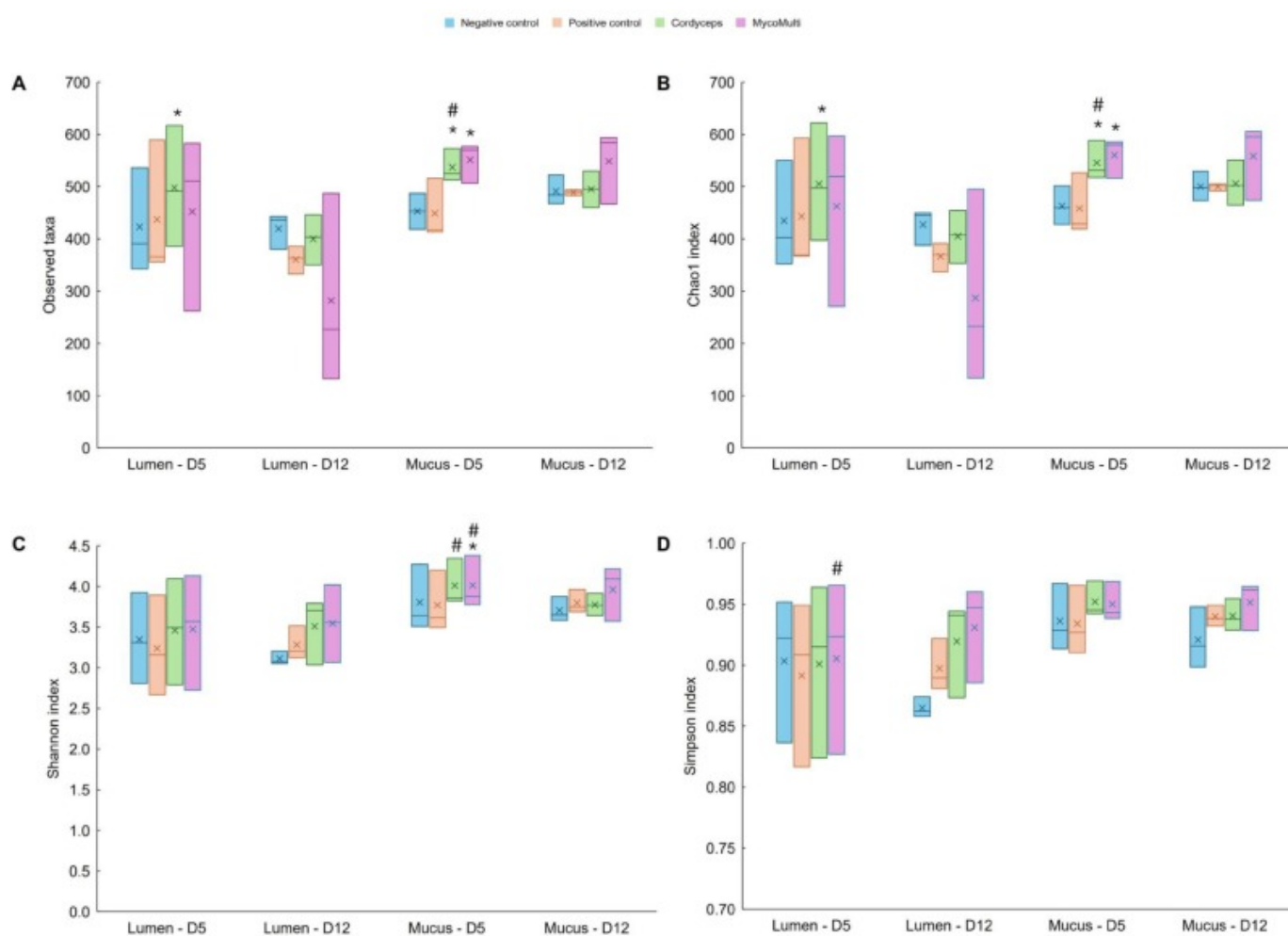
[Download: Download high-res image \(399KB\)](#)
[Download: Download full-size image](#)

Fig. 3. OPLS-DA-score plot based on the LA-REIMS data for all experimental conditions. LA-REIMS was used to measure the metabolic activity of colonic fermentations collected from the Screening-SHIME® experiment at D12 using fecal samples from three healthy donors. CV-ANOVA, cross-validation analysis of variance; D, day; LA-REIMS, Laser-Assisted Rapid Evaporative Ionization Mass Spectrometry; OPLS-DA, Orthogonal Partial Least Squares Discriminant Analysis; SHIME®, Simulator of the Human Intestinal Microbial Ecosystem.

3.3. Metagenomic analysis

3.3.1. Alpha diversity

Results for the alpha-diversity analysis are shown in Fig. 4. As measures of species richness, both the Observed taxa and Chao1 indices demonstrated a significant increase in alpha-diversity at D5 with M2 Cordyceps 116™ supplementation in the lumen environment and with M2 Cordyceps 116™ and Myco-Multi™ supplementation in the mucus environment versus the negative control, and with M2 Cordyceps 116™ in the mucus environment versus the positive control (Fig. 4A and B). With the Shannon index, there was a significant increase in alpha-diversity with Myco-Multi™ supplementation versus the negative control and with both M2 Cordyceps 116™ and Myco-Multi™ supplementation versus the positive control in the mucosal environment at D5 (Fig. 4C). Using the Simpson index, there was a significant increase in alpha-diversity with Myco-Multi™ supplementation versus the positive control in the lumen environment at D5 (Fig. 4D). There were no significant differences between the negative or positive controls and the mushroom test products at D12 with any of the alpha-diversity indices.



[Download: Download high-res image \(391KB\)](#)

[Download: Download full-size image](#)

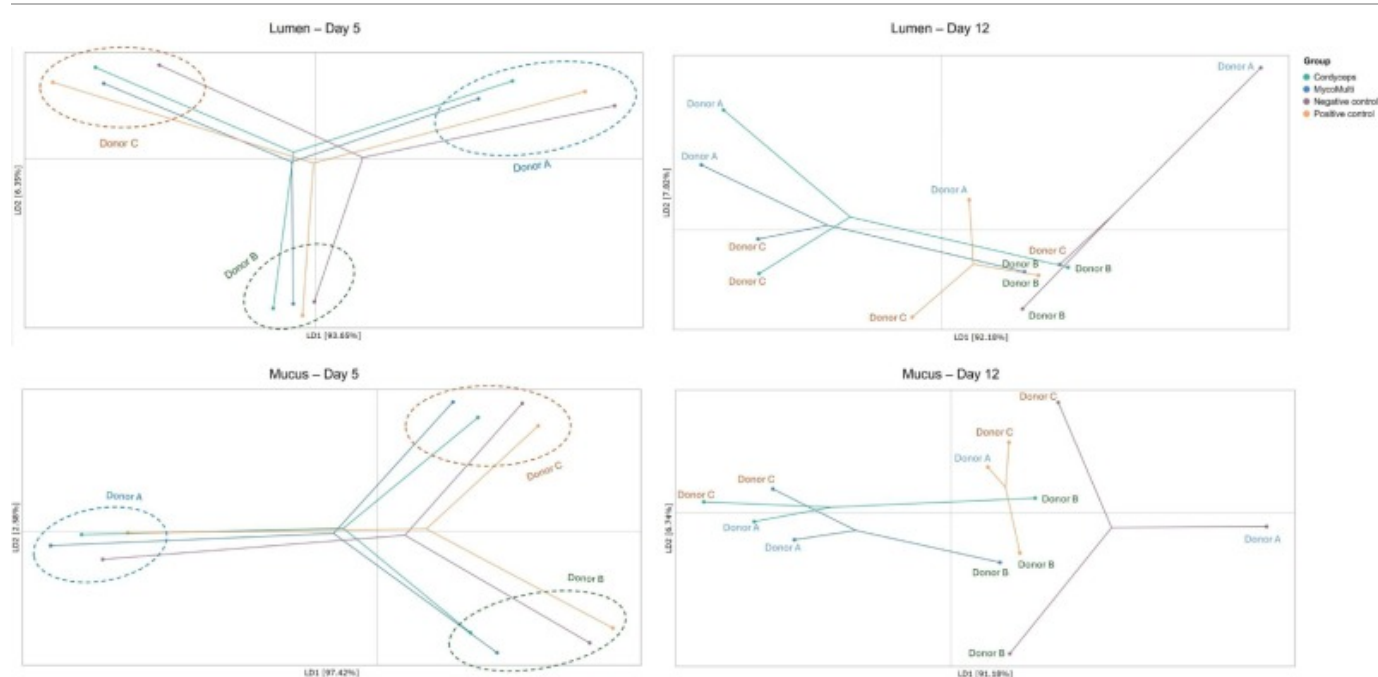
Fig. 4. Boxplots showing the alpha-diversity across donors at Day 5 and Day 12 for the Observed taxa index (A), the Chao1 index (B), the Shannon index (C), and the Simpson index (D). Alpha-diversity data at D5 and D12 in the lumen and mucosal environments of the Screening-SHIME® model using fecal samples from three healthy human donors. The × represents the average across

donors and the horizontal line represents the median across donors ($n = 3$). Paired two-tailed student's t-tests were used for statistical comparisons. * $p < 0.05$ versus the negative control. # $p < 0.05$ versus the positive control.

D, day; SHIME®, Simulator of the Human Intestinal Microbial Ecosystem.

3.3.2. Beta diversity

In both the lumen and mucosal environments, the microbial communities were clustered by donor at D5 followed by a clear separation between the negative control and the two mushroom products, particularly for donors A and C, at D12 (Fig. 5). The effect was somewhat greater with M2 Cordyceps 116™ than Myco-Multi™. The difference between the negative and positive controls was less substantial than the difference between the negative control and the mushroom test products.



[Download: Download high-res image \(355KB\)](#)

[Download: Download full-size image](#)

Fig. 5. Discriminant Analysis of Principal Component (DAPC) plots representing beta-diversity.

Beta-diversity data at D5 and D12 in the lumen and mucosal environments of the Screening-SHIME® model using fecal samples from three healthy human donors (A, B, and C).

D, day; SHIME®, Simulator of the Human Intestinal Microbial Ecosystem.

3.3.3. Differential abundance analysis

There was a notable decrease in the relative abundance of *Bilophila wadsworthia* with both M2 Cordyceps 116™ and Myco-Multi™ supplementation compared with the negative and positive controls in both the luminal and mucosal environments (Table 1, Table 2, S1, and S2).

Table 1. Overview of supplementation-induced enrichments compared with the negative control in bacterial species in the lumen.

Phylum	Family	Genus	Species	Day 5			Day 12		
				PC	CC	MM	PC	CC	MM
Actinomycetota	Bifidobacteriaceae	Bifidobacterium	<i>Bifidobacterium adolescentis</i>				+	+	
			<i>Bifidobacterium catenulatum</i>				+		
			<i>Bifidobacterium ruminantium</i>					+	
Bacillota	Bacillaceae_G	Bacillus_A	<i>Bacillus_A mobilis</i>	+	+	+			
			<i>Bacillus_A thuringiensis</i>		+				
	DSM-18226	Cytobacillus	<i>Cytobacillus horneckiae</i>			+			
	Paenibacillaceae	Paenibacillus	<i>Paenibacillus etheri</i>				-	-	
			<i>Paenibacillus sp000499205</i>	+			+	+	
			<i>Paenibacillus sp000758545</i>				-	-	
			<i>Paenibacillus sp000758585</i>				-	-	
			<i>Paenibacillus sp002264395</i>				-	-	
		-	-		+				
	Planococcaceae	<i>Lysinibacillus</i>	<i>Lysinibacillus fusiformis</i>					-	
		<i>Viridibacillus</i>	<i>Viridibacillus arvi</i>	+			+		
Bacillota_A	CAG-138	RUG472	RUG472 sp900545265				+	+	
	Clostridiaceae	Clostridium	<i>Clostridium tertium</i>			+			
			<i>Agathobacter faecis</i>					+	+
	Lachnospiraceae	<i>Anaerostipes</i>	<i>Anaerostipes sp900756035</i>				-		-
		Blautia_A	<i>Blautia_A faecis</i>				+	+	+
			<i>Blautia_A sp905206915</i>				+	+	+
		CAG-317	CAG-317 sp000433215						+
		<i>Choladocola</i>	<i>Choladocola sp003480725</i>					+	
		<i>Clostridium_Q</i>	<i>Clostridium_Q fessum</i>				-		
		<i>Dorea_A</i>	<i>Dorea_A longicatena</i>				+		
		<i>Enterocloster</i>	<i>Enterocloster clostridioformis_A</i>				-		
		<i>Eubacterium_G</i>	<i>Eubacterium_G sp000434315</i>					+	
		<i>Lachnoclostridium_B</i>	<i>Lachnoclostridium_B sp900066555</i>					+	
		SFDP01	SFDP01 sp004558185						-
		<i>Roseburia</i>	<i>Roseburia hominis</i>					+	
	Oscillospiraceae	<i>Flavonifractor</i>	<i>Flavonifractor plautii</i>						-
	Ruminococcaceae	Faecalibacterium	<i>Faecalibacterium prausnitzii</i>					+	
			<i>Faecalibacterium prausnitzii_D</i>					+	
			<i>Faecalibacterium sp900758465</i>						+
		<i>Gemmiger</i>	<i>Gemmiger sp900540595</i>						+

Bacteroidota	Bacteroidaceae	Bacteroides	<i>Bacteroides caccae</i>					-	-
			<i>Bacteroides eggerthii</i>		+				
			<i>Bacteroides faecis</i>	-					
			<i>Bacteroides sp900556625</i>						-
			<i>Bacteroides sp900761785</i>	-	-	-			
			<i>Bacteroides thetaiotaomicron</i>	-	-				
	Tannerellaceae	<i>Paraprevotella</i>	<i>Paraprevotella clara</i>						+
		<i>Phocaeicola</i>	<i>Phocaeicola dorei</i>						+
		<i>Parabacteroides</i>	<i>Parabacteroides merdae</i>					-	
Desulfobacterota	Desulfovibrionaceae	<i>Bilophila</i>	<i>Bilophila wadsworthia</i>						-
		<i>Desulfovibrio</i>	<i>Desulfovibrio piger_A</i>						+
		<i>Mailhella</i>	-						-
Pseudomonadota	<i>Burkholderiaceae_C</i>	<i>Achromobacter</i>	<i>Achromobacter xylosoxidans</i>		+	+			
	<i>Enterobacteriaceae</i>	<i>Kosakonia</i>	<i>Kosakonia sp018261615</i>				-		-
		<i>Phytobacter</i>	<i>Phytobacter diazotrophicus</i>	-	-	-	-	-	-
			<i>Phytobacter palmae</i>			-	-		
			<i>Phytobacter ursingii</i>			-	-		
	<i>Pseudomonadaceae</i>	<i>Pseudomonas</i>	<i>Pseudomonas Aeruginosa</i>	+		+	+		+
			<i>Pseudomonas Aeruginosa_A</i>			+			+
	<i>Xanthomonadaceae</i>	<i>Stenotrophomonas</i>	<i>Stenotrophomonas maltophilia_S</i>			+		-	

Enrichments were identified with LEfSe and/or treeclimbR. Colors indicate the type of significance. Red shading indicates both statistical and biological significance ($p < 0.05$, $FC > 4$) and blue shading indicates statistical significance ($p < 0.05$, $FC < 4$).

CC, M2 Cordyceps 116™; FC, fold change; LEfSe, Linear Discriminant Analysis Effect Size; MM, Myco-Multi™; PC, positive control.

Table 2. Overview of supplementation-induced enrichments compared with the negative control in bacterial species in the mucus.

Phylum	Family	Genus	Species	Day 5			Day 12		
				PC	CC	MM	PC	CC	MM
Actinomycetota	Bifidobacteriaceae	Bifidobacterium	Bifidobacterium adolescentis				+		+
	Coriobacteriaceae	Collinsella	Collinsella sp900553165						+
Bacillota_A	Clostridiaceae	Clostridium	Clostridium tertium			+			
		Clostridium_Q	clostridium Q symbiosum					-	-
			Clostridium_Q fessum				-	-	-
		Copromonas	Copromonas sp900066535				-		
		Dorea_A	Dorea_A longicatena				+		
		Enterocloster	Enterocloster bolteae				-		
		Eubacterium_G	Eubacterium_G sp000434315					+	+
		KLE1615	KLE1615 sp900066985				+		
		Mediterraneibacter	Mediterraneibacter lactaris						+
	Oscillospiraceae	Dysosmobacter	Dysosmobacter welbionis						+
		Lawsonibacter	Lawsonibacter sp900754605				-		
	Ruminococcaceae	Bittarella	Bittarella massiliensis			+			
		Gemmiger	Gemmiger qucibialis					+	
Bacillota_C	Acidaminococcaceae	Acidaminococcus	Acidaminococcus					+	
			sp900538365						
Bacteroidota	Bacteroidaceae	Bacteroides	Bacteroides thetaiotaomicron						-
		Paraprevotella	Paraprevotella clara						+
			Paraprevotella xylaniphila						+
	Tannerellaceae	Parabacteroides	Parabacteroides merdae				-		
Desulfobacterota	Desulfobacteriaceae	Bilophila	Bilophila wadsworthia					-	-
Pseudomonadota	Burkholderiaceae_A	Sutterella	Sutterella wadsworthensis				+	+	+
	Burkholderiaceae_C	Achromobacter	Achromobacter xylosoxidans		+	+			
	Enterobacteriaceae	Phytobacter	Phytobacter diazotrophicus	-	-	-	-	-	-
	Pseudomonadaceae	Pseudomonas	Pseudomonas Aeruginosa			+	+		+
	Xanthomonadaceae	Stenotrophomonas	Stenotrophomonas maltophilia_S			+		-	

Enrichments were identified with LEfSe and/or treeclimbR. Colors indicate the type of significance. Red shading indicates both statistical and biological significance ($p < 0.05$, $FC > 4$) and blue shading indicates statistical significance ($p < 0.05$, $FC < 4$).

CC, M2 Cordyceps 116™; FC, fold change; LEfSe, Linear Discriminant Analysis Effect Size; MM, Myco-Multi™; PC, positive control.

In the luminal environment, M2 Cordyceps 116™ supplementation significantly enriched *Bifidobacterium* spp., *Lachnospiraceae* members such as *Blautia*_A spp., *Lachnoclostridium*_B sp900066555, *Choladocola* sp003480725, *Agathobacter faecis*, *Eubacterium_G* sp000434315, and *Roseburia hominis* (Table 1). Furthermore, known butyrate producers *Faecalibacterium prausnitzii* and *Faecalibacterium prausnitzii_D* (Barcenilla et al., 2000; Duncan et al., 2002) were enriched following M2 Cordyceps 116™ supplementation. In the mucosal environment, *Eubacterium_G* sp000434315 and *Gemmiger qucibialis* were enriched (Table 2). Compared with the positive control, M2 Cordyceps 116™ consistently enhanced beneficial bacterial populations in the luminal

environment across multiple donors, including butyrate producers *Eubacterium_G sp000434315*, *Roseburia hominis*, and *Faecalibacterium prausnitzii_D* (**Table S1**) (Barcenilla et al., 2000; Duncan et al., 2002; Litty, et al., 2021; Song et al., 2022). In the mucosal environment, relative abundances of members of the Bacillota_A phylum such as *Eubacterium_G sp000434315*, *Gemmiger qucibialis*, and *Dysosmobacter* genera were enriched with M2 Cordyceps 116™ supplementation (**Table S2**).

In the luminal environment, Myco-Multi™ supplementation enriched members of the *Lachnospiraceae* family such as *Agathobacter faecis* and *Blautia_A* species (**Table 1**). Butyrate producers, such as *Faecalibacterium sp900758465* and *Gemmiger sp900540595* (Barcenilla et al., 2000; Duncan et al., 2002; L. Zhang, et al., 2025) were enriched, as well as *Paraprevotella clara* and *Phocaeicola dorei* (*Bacteroidaceae* family). In the mucosal environment, Myco-Multi™ supplementation increased Actinomycetota members *Bifidobacterium adolescentis* and *Collinsella sp900553165*, Bacillota_A members *Eubacterium_G sp000434315* and *Mediterraneibacter lactaris* (*Lachnospiraceae* family), and *Dysosmobacter welbionis* (*Oscillospiraceae* family) (**Table 2**). *Paraprevotella* spp. were also enriched. Compared with the positive control, Myco-Multi™ supplementation resulted in enrichment of butyrate producers *Faecalibacterium sp900758465* and *Gemmiger sp90054059* (*Ruminococcaceae* family) and *Paraprevotella* spp. in the luminal environment (**Table S1**). In the mucosal environment, Myco-Multi™ supplementation increased the abundance of *Bacillus_A mobilis*, *Eubacterium* spp., and *Dysosmobacter* spp. (**Table S2**). The growth of *Bacteroides eggerthii* and *Paraprevotella* spp. were also enhanced.

4. Discussion

This study evaluated the effects of two functional whole mushroom powders (M2 Cordyceps 116™ and Myco-Multi™) on the gut microbiota of three individual human donors using the in vitro Screening-SHIME® model. Both test products demonstrated a prebiotic effect, with increased gas pressure and SCFA production compared with the negative control, and distinct bacterial enrichments after repeated supplementation. Further, the two test products had different metabolomic fingerprints, suggesting that beta-glucans and chitin were not the only drivers of the changes observed with test product supplementation. Metagenomic analysis demonstrated enrichments of beneficial bacterial species and a decrease in some pathogenic species following supplementation of both test products.

Gas production, indicated by gas pressure in this study, represents overall microbial fermentative activity. Fermentative activity was demonstrated for inulin, M2 Cordyceps 116™, and Myco-Multi™, with significant increases versus the negative control for inulin and M2 Cordyceps 116™. The increase in gas pressure may have benefitted gas-consuming, acetate-producing *Blautia_A* spp. (Mutuyemungu et al., 2023), which were enhanced with both mushroom test products. Overall microbial activity was similar (i.e., no significant differences) between the test products and inulin, supporting a prebiotic role for the two test products of interest.

By D8, M2 Cordyceps 116™ demonstrated a strong butyrogenic effect compared with inulin. This was sustained through D12, but did not reach significance on D11 and D12 due to donor variability. This may be linked with an increase in the relative abundance of several butyrate producing bacterial species with M2 Cordyceps 116™ supplementation, including *Faecalibacterium prausnitzii*, *Faecalibacterium prausnitzii_D*, and *Eubacterium_G sp000434315* (Barcenilla et al., 2000; Duncan, Holtrop, et al., 2004; Litty, et al., 2021). While the difference in butyrate levels between inulin and Myco-Multi™ did not reach statistical significance, there was a numerical increase in the mean and median concentration of butyrate at D8, and even more so at D11 and D12 with Myco-Multi™ supplementation. This may also be linked with an increase in the relative abundance of butyrate-producing bacterial species, such as *Gemmiger sp900540595*, *Faecalibacterium sp900758465* (Barcenilla et al., 2000; Duncan, Holtrop, et al., 2004; L. Zhang, et al., 2025). Further, for both mushroom test products, lactate levels increased at D3, then decreased to nearly undetectable levels following supplementation (M2 Cordyceps 116™, D8, D11, and D12; Myco-Multi™, D5, D8, D11, and D12). Some bacteria utilize lactate, converting it to butyrate in a process called cross-feeding (Duncan, Louis, et al., 2004; Rios-Covian et al., 2015). The decrease in lactate concentrations in parallel with the increase in butyrate concentrations suggest cross-feeding, which is supported by the enrichment of lactate producers *Bifidobacterium adolescentis* and *Bifidobacterium rumenatum* with M2 Cordyceps 116™ supplementation and *Bifidobacterium adolescentis* with Myco-Multi™ supplementation, and the increased absolute abundance of cross-feeding butyrate producers such as *Faecalibacterium* spp. (Rios-Covian et al., 2015). The butyrogenic effects of the mushroom test products, which was stronger than that observed with inulin, highlights their prebiotic activity. Furthermore, the stimulation of several of these bacterial species has been linked with beneficial effects for the host. *Bifidobacterium adolescentis*, for instance, is known for the production of folate (next to acetate and lactate), a characteristic which has been linked to its probiotic properties (Pompei et al., 2007). Moreover, attenuation of *Bifidobacterium adolescentis* levels have been found in patients suffering from inflammatory bowel disease (Joossens et al., 2011) as well as in allergic individuals (Hevia et al., 2016). The butyrate-producing *Faecalibacterium prausnitzii*, on the other hand, exerts strong anti-inflammatory activity resulting in decreased inflammatory markers in the intestinal environment (Furet et al., 2010), and has also been shown to reduce endotoxemia in obese subjects (González-Sarrías et al., 2018).

Untargeted metabolomics revealed a distinct clustering for each of the mushroom test products and the positive and negative controls on D12. The distinct metabolomic profiles for M2 Cordyceps 116™ and Myco-Multi™ supplementation indicate that the changes in microbial metabolite production cannot be fully explained by the activity related to digestion of the main fiber compounds of mushrooms, i.e., beta-glucan and chitin, alone. Future studies utilizing ultra-high performance liquid chromatography-high-resolution mass spectrometry system (UHPLC-HRMS) to identify changes in specific metabolites following M2 Cordyceps 116™ and Myco-Multi™ supplementation would elucidate the differentiating metabolites stimulated by the two products, which could be of particular interest.

Overall, the changes in species richness and/or evenness (alpha diversity) with test product supplementation were mild. While some significant changes in both species richness and evenness were reported with the test products compared with the negative control at D5, they were generally diminished by D12 of repeated supplementation, suggesting a return to bacterial homeostasis within the colonic environment. This is considered a positive effect ([Lee et al., 2022](#)), demonstrating that both M2 Cordyceps 116™ and Myco-Multi™ support homeostasis of the healthy gut microbiome.

Metagenomics results showed a decrease for some pathogenic bacterial species and enrichments of several beneficial bacterial species with the test products versus both the negative and positive controls. The metagenomic changes were more pronounced on D12 than on D5, suggesting the importance of repeated administration of the test products. The test products affected the relative abundance of some bacterial species that may be considered pathogens. For example, supplementation with either M2 Cordyceps 116™ (mucosal environment) or Myco-Multi™ (luminal and mucosal environments) significantly decreased the relative abundance of *Bilophila wadsworthia*, a sulfur dioxide producing commensal bacteria, compared with the negative control ([Waqas et al., 2023](#)). The relative abundance of *Bilophila wadsworthia* was also significantly decreased compared with inulin (M2 Cordyceps 116™, mucosal environment; Myco-Multi™, luminal environment). Sulfur dioxide can damage the mucosal lining of the colon and may contribute to the development of colorectal cancer and inflammatory bowel disease ([Waqas et al., 2023](#)). Additionally, it is reported to be associated with an aggravation of high fat diet-induced metabolic dysfunction (e.g., intestinal barrier dysfunction, inflammation) in mice ([Natividad et al., 2018](#)). Thus, a reduction is considered beneficial.

A number of SCFA-producing bacterial species were enriched with M2 Cordyceps 116™ or Myco-Multi™ versus the negative control and versus inulin (positive control) in the luminal and mucosal environments. The bacterial species that were enriched differed somewhat between the two mushroom test products, further supporting that the two products support a distinct set of changes in the gut microbiome. SCFAs are an important contribution of the gut microbiome to human health. Butyrate is of particular importance as it is a primary source of energy for intestinal epithelial cells and inhibits the expansion of cancer cells. SCFAs in general play an important role and have been associated with increased circulation of gut hormones such as glucagon-like peptide-1 (GLP-1), serotonin, and peptide YY (PYY) ([Martin-Gallausiaux et al., 2021](#); [Mazhar et al., 2023](#)), which are involved in regulation of sugar metabolism and appetite ([Muller et al., 2019](#)); modulation of the enteric nervous system (e.g., intestinal motility, inflammation) ([Terry, et al., 2017](#)); and electrolyte absorption and appetite regulation ([Batterham, et al., 2003](#); [McCauley et al., 2020](#)), respectively. Despite their important role in metabolism and appetite control, colonic levels of GLP-1 and PYY decline with age in humans ([Jones et al., 2023](#)). The stimulation of increased SCFA/butyrate production with M2 Cordyceps 116™ and Myco-Multi™ supplementation can thus be considered a beneficial effect, considering their effects on intestinal epithelial cell health and gut hormone regulation.

This study had a few limitations. First, this study utilized fecal samples from three individual donors which allowed for some investigation into interindividual differences; however, the small number of donors limits the statistical power of the analyses and the generalizability of the results. Second, the study findings were obtained using an in vitro model. While this model has been validated with in vivo parameters (Molly et al., 1993) and allows for greater exploration into the mechanistic aspects of the observed prebiotic effects of the test products than would be feasible or ethical in an in vivo study, the results cannot be directly correlated to a biological response in vivo. As such, confirmatory studies in humans are needed. However, it should be mentioned that the used SHIME® model has already shown strong correlation between in vitro and in vivo data with respect to prebiotic metabolism. For example, the effects of repeated inulin administration in a clinical validation study (Bouhnik, et al., 2007) confirmed the findings obtained during an in vitro SHIME® study (Van de Wiele et al., 2004), as shown by a strong butyrogenic effect and specific stimulation of the bifidobacterial population.

5. Conclusion

Overall, repeated supplementation of either M2 Cordyceps 116™ or Myco-Multi™ resulted in increased microbial metabolic activity, including an increase in SCFAs, particularly butyrate, compared with both a negative control and the established prebiotic inulin. Further, supplementation of both test products resulted in distinct changes in the microbial metabolome and in the microbial community composition, indicating not only their prebiotic activity, but also suggesting that these effects were driven by more than just beta-glucans and chitin. Functional whole mushroom ingredients could thus be an added value in the human diet to support gut homeostasis and maintain healthy gut microbiome function.

CRedit authorship contribution statement

Julie Daoust: Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Jay Schmalz:** Writing – review & editing, Funding acquisition, Conceptualization. **Chloë Rotsaert:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Chelsea Vannieuwenhuyse:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Jelle De Medts:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Lieven Van Meulebroek:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis, Data curation. **Cindy Duysburgh:** Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Data curation.

Ethics statement

Fecal materials were collected and used as approved by the Ethics Committee of the University Hospital Ghent (reference number ONZ-2022-0267; approved on 29 July 2022).

Financial support

This study was funded by M2 Ingredients. The study sponsor was involved in the study design; in the writing of the report; and in the decision to submit the article for publication. The study sponsor was not involved in the collection, analysis, and interpretation of data.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: The study was financially supported by M2 Ingredients. Julie Daoust and Jay Schmalz are employees of M2 Ingredients. Other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors thank Sarah Bubeck, PhD, of Bubeck Scientific Communications for providing medical writing support.

Appendix A. Supplementary data

 [Download: Download Word document \(29KB\)](#)

Supplementary material.

[Recommended articles](#)

Data availability

Data will be made available on request.

References

[Araujo-Rodrigues, Sousa, Relvas, Tavaría and Pintado, 2024](#) H. Araujo-Rodrigues, A.S. Sousa, J.B. Relvas, F.K. Tavaría, M. Pintado

An overview on mushroom polysaccharides: Health-promoting properties, prebiotic and gut microbiota modulation effects and structure-function correlation

Carbohydrate Polymers, 333 (2024), Article 121978, [10.1016/j.carbpol.2024.121978](https://doi.org/10.1016/j.carbpol.2024.121978) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Atik, Atik, Akarca and Denizkara, 2024](#) A. Atik, I. Atik, G. Akarca, A.J. Denizkara

Incorporating Ganoderma lucidum extract and powder with probiotic cultures

(*Lactobacillus acidophilus* and *Bifidobacterium animalis* subsp. *lactis*) enhanced the functional, textural, and sensory qualities of yogurt

Journal of Food Science and Technology (2024), [10.1007/s13197-024-06068-z](https://doi.org/10.1007/s13197-024-06068-z) ↗

[Google Scholar](#) ↗

[Bai et al., 2024](#) M. Bai, X. Wang, D. Liu, A. Xu, H. Cheng, L. Li, C. Zhang

Tolypocladium sinense mycelium polysaccharide alleviates obesity, lipid metabolism disorder, and inflammation caused by high fat diet via improving intestinal barrier and modulating gut microbiota

Molecular Nutrition & Food Research, 68 (9) (2024), Article e2300759, [10.1002/mnfr.202300759](https://doi.org/10.1002/mnfr.202300759) ↗

[Google Scholar](#) ↗

[Barcenilla et al., 2000](#) A. Barcenilla, S.E. Pryde, J.C. Martin, S.H. Duncan, C.S. Stewart, C. Henderson, H.J. Flint

Phylogenetic relationships of butyrate-producing bacteria from the human gut

Applied and Environmental Microbiology, 66 (4) (2000), pp. 1654-1661, [10.1128/AEM.66.4.1654-1661.2000](https://doi.org/10.1128/AEM.66.4.1654-1661.2000) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Batterham and Bloom, 2003](#) R.L. Batterham, S.R. Bloom

The gut hormone peptide YY regulates appetite

Annals of the New York Academy of Sciences, 994 (2003), pp. 162-168, [10.1111/j.1749-6632.2003.tb03176.x](https://doi.org/10.1111/j.1749-6632.2003.tb03176.x) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Bouchnik et al., 2007](#) Y. Bouchnik, L. Raskine, K. Champion, C. Andrieux, S. Penven, H. Jacobs, G. Simoneau

Prolonged administration of low-dose inulin stimulates the growth of bifidobacteria in humans

Nutrition Research, 27 (4) (2007), pp. 187-193, [10.1016/j.nutres.2007.01.013](https://doi.org/10.1016/j.nutres.2007.01.013) ↗

[Google Scholar](#) ↗

[Carlson, Erickson, Lloyd and Slavin, 2018](#) J.L. Carlson, J.M. Erickson, B.B. Lloyd, J.L. Slavin

Health effects and sources of prebiotic dietary fiber

Current Developments in Nutrition, 2 (3) (2018), Article nzy005, [10.1093/cdn/nzy005](https://doi.org/10.1093/cdn/nzy005) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Chrysostomou et al., 2024](#) P.P. Chrysostomou, E. Freeman, M.M. Murphy, A. Chaudhary, N. Siddiqui, J. Daoust

A toxicological assessment of *Ganoderma lucidum* and *Cordyceps militaris* mushroom powders

Frontiers in Toxicology (Food and Nutritional Toxicology Section), 6 (2024), p. 1469348, [10.3389/ftox.2024.146934](https://doi.org/10.3389/ftox.2024.146934) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

Coherent Market Insights. Global Medicinal Mushroom Market Size and Share Analysis – Growth Trends and Forecasts, 2025–2032

Coherent Market Insights. Global Medicinal Mushroom Market Size and Share Analysis – Growth Trends and Forecasts (2025–2032). Published February 2025. Accessed September 4, 2025. <https://www.coherentmarketinsights.com/industry-reports/global-medicinal-mushroom-market> ↗

[Google Scholar](#) ↗

Cortes Bordoy et al., 2023 J. Cortes Bordoy, J. de Santiago Garcia, M. Agenjo Gonzalez, D. Dexeus Carter, G. Fiol Ruiz, C. Garcia Ferreiro, ..., A.E. Del Villar Vazquez

Effect of a multi-ingredient Coriolus-versicolor-based vaginal gel in women with HPV-dependent cervical lesions: The Papilobs real-life prospective study

Cancers (Basel), 15 (15) (2023), p. 3863, [10.3390/cancers15153863](https://doi.org/10.3390/cancers15153863) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

Davani-Davari et al., 2019 D. Davani-Davari, M. Negahdaripour, I. Karimzadeh, M. Seifan, M. Mohkam, S.J. Masoumi, ..., Y. Ghasemi

Prebiotics: Definition, types, sources, mechanisms, and clinical applications

Foods, 8 (3) (2019), p. 92, [10.3390/foods8030092](https://doi.org/10.3390/foods8030092) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

De Weirdt et al., 2010 R. De Weirdt, S. Possemiers, G. Vermeulen, T.C. Moerdijk-Poortvliet, H.T. Boschker, W. Verstraete, T. Van de Wiele

Human faecal microbiota display variable patterns of glycerol metabolism

FEMS Microbiology Ecology, 74 (3) (2010), pp. 601-611, [10.1111/j.1574-6941.2010.00974.x](https://doi.org/10.1111/j.1574-6941.2010.00974.x) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

Deehan et al., 2024 E.C. Deehan, S. Al Antwan, R.S. Witwer, P. Guerra, T. John, L. Monheit

Revisiting the concepts of prebiotic and prebiotic effect in light of scientific and regulatory progress-A consensus paper from the global prebiotic association

Advances in Nutrition, 15 (12) (2024), Article 100329, [10.1016/j.advnut.2024.100329](https://doi.org/10.1016/j.advnut.2024.100329) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

Diling et al., 2017 C. Diling, Z. Chaoqun, Y. Jian, L. Jian, S. Jiyan, X. Yizhen, L. Guoxiao

Immunomodulatory activities of a fungal protein extracted from Hericium erinaceus through regulating the gut microbiota

Frontiers in Immunology, 8 (2017), p. 666, [10.3389/fimmu.2017.00666](https://doi.org/10.3389/fimmu.2017.00666) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

Duncan, Hold, Harmsen, Stewart and Flint, 2002 S.H. Duncan, G.L. Hold, H.J.M. Harmsen, C.S. Stewart, H.J. Flint

Growth requirements and fermentation products of fusobacterium prausnitzii, and

a proposal to reclassify it as *Faecalibacterium prausnitzii* gen. Nov., comb. nov

International Journal of Systematic and Evolutionary Microbiology, 52 (Pt 6) (2002), pp. 2141-2146, [10.1099/00207713-52-6-2141](https://doi.org/10.1099/00207713-52-6-2141) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Duncan et al., 2004](#) S.H. Duncan, G. Holtrop, G.E. Lobley, A.G. Calder, C.S. Stewart, H.J. Flint

Contribution of acetate to butyrate formation by human faecal bacteria

The British Journal of Nutrition, 91 (6) (2004), pp. 915-923, [10.1079/BJN20041150](https://doi.org/10.1079/BJN20041150) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Duncan, Louis and Flint, 2004](#) S.H. Duncan, P. Louis, H.J. Flint

Lactate-utilizing bacteria, isolated from human feces, that produce butyrate as a major fermentation product

Applied and Environmental Microbiology, 70 (10) (2004), pp. 5810-5817, [10.1128/AEM.70.10.5810-5817.2004](https://doi.org/10.1128/AEM.70.10.5810-5817.2004) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Duysburgh et al., 2024](#) C. Duysburgh, L. Miclotte, J.B. Green, K.T. Watts, M.I. Sardi, A. Chakrabarti, ..., M. Marzorati

Saccharomyces cerevisiae derived postbiotic alters gut microbiome metabolism in the human distal colon resulting in immunomodulatory potential in vitro

Frontiers in Microbiology, 15 (2024), Article 1358456, [10.3389/fmicb.2024.1358456](https://doi.org/10.3389/fmicb.2024.1358456) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Furet et al., 2010](#) J.P. Furet, L.C. Kong, J. Tap, C. Poitou, A. Basdevant, J.L. Bouillot, ..., K. Clément

Differential adaptation of human gut microbiota to bariatric surgery-induced weight loss: Links with metabolic and low-grade inflammation markers

Diabetes, 59 (2010), pp. 3049-3057, [10.2337/db10-0253](https://doi.org/10.2337/db10-0253) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Gamage, Nakayama, Fuyuno and Ohga, 2018](#) S. Gamage, J. Nakayama, Y. Fuyuno, S. Ohga

The effect of the hot water extracts of the *Paecilomyces hepiali* and *Cordyceps militaris* mycelia on the growth of gastrointestinal bacteria

Advances in Microbiology, 8 (7) (2018), pp. 490-505, [10.4236/aim.2018.87034](https://doi.org/10.4236/aim.2018.87034) ↗

[Google Scholar ↗](#)

[González-Sarrías, Romo-Vaquero, Carcía-Villalba and A., Selma, M.V., & Espín, J.C., 2018](#) A. González-Sarrías, M. Romo-Vaquero, C.-M. Carcía-Villalba, A., Selma, M.V., & Espín, J.C.

The endotoxemia marker lipopolysaccharide-binding protein is reduced in overweight-obese subjects consuming pomegranate extract by modulating the gut microbiota: A randomized clinical trial

Molecular Nutrition & Food Research, 62 (2018), Article e1800160, [10.1002/mnfr.201800160](https://doi.org/10.1002/mnfr.201800160) ↗

[Google Scholar ↗](#)

[Hess, Wang, Gould and Slavin, 2018](#) J. Hess, Q. Wang, T. Gould, J. Slavin

Impact of Agaricus bisporus mushroom consumption on gut health markers in healthy adults

Nutrients, 10 (10) (2018), p. 1402, [10.3390/nu10101402 ↗](#)

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Hevia et al., 2016](#) A. Hevia, C. Milani, P. López, C.D. Donado, A. Cuervo, S. González, A. Suárez, F. Turróni, M. Gueimonde, M. Ventura, B. Sánchez, A. Margolles

Allergic patients with long-term asthma display low levels of Bifidobacterium adolescentis

PLoS One, 11 (2) (2016), Article e0147809, [10.1371/journal.pone.0147809 ↗](#)

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Hoefman, Pommerening-Roser, Samyn, De Vos and Heylen, 2013](#) S. Hoefman, A. Pommerening-Roser, E. Samyn, P. De Vos, K. Heylen

Efficient cryopreservation protocol enables accessibility of a broad range of ammonia-oxidizing bacteria for the scientific community

Research in Microbiology, 164 (4) (2013), pp. 288-292, [10.1016/j.resmic.2013.01.007 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[Huang et al., 2021](#) R. Huang, C. Soneson, P.L. Germain, T.S.B. Schmidt, C.V. Mering, M.D. Robinson

treeclimbR pinpoints the data-dependent resolution of hierarchical hypotheses

Genome Biology, 22 (1) (2021), p. 157, [10.1186/s13059-021-02368-1 ↗](#)

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Hughes, Alvarado, Swanson and Holscher, 2022](#) R.L. Hughes, D.A. Alvarado, K.S. Swanson, H.D. Holscher

The prebiotic potential of inulin-type fructans: A systematic review

Advances in Nutrition, 13 (2) (2022), pp. 492-529, [10.1093/advances/nmab119 ↗](#)

 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

[Jombart, 2008](#) T. Jombart

Adegenet: A R package for the multivariate analysis of genetic markers

Bioinformatics, 24 (11) (2008), pp. 1403-1405, [10.1093/bioinformatics/btn129 ↗](#)

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Jombart, Devillard and Balloux, 2010](#) T. Jombart, S. Devillard, F. Balloux

Discriminant analysis of principal components: A new method for the analysis of genetically structured populations

BMC Genetics, 11 (2010), p. 94, [10.1186/1471-2156-11-94 ↗](#)

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Jones et al., 2023](#) L.A. Jones, E.W. Sun, A.L. Lumsden, D.W. Thorpe, R.A. Peterson, D. De Fontgalland, ..., D.J.

Keating

Alterations in GLP-1 and PYY release with aging and body mass in the human gut

Molecular and Cellular Endocrinology, 578 (2023), Article 112072, [10.1016/j.mce.2023.112072](#) ↗



[View PDF](#)

[View article](#)

[View in Scopus](#) ↗

[Google Scholar](#) ↗

[Joossens et al., 2011](#) M. Joossens, G. Huys, M. Cnockaert, V. De Preter, K. Verbeke, P. Rutgeerts, P. Vandamme, S. Vermeire

Dysbiosis of the faecal microbiota in patients with Crohn's disease and their unaffected relatives

Gut, 60 (2011), pp. 631-637, [10.1136/gut.2010.223263](#) ↗

[View in Scopus](#) ↗

[Google Scholar](#) ↗

[Lee, Tsolis and Baumler, 2022](#) J.Y. Lee, R.M. Tsolis, A.J. Baumler

The microbiome and gut homeostasis

Science, 377 (6601) (2022), Article eabp9960, [10.1126/science.abp9960](#) ↗

[View in Scopus](#) ↗

[Google Scholar](#) ↗

[Litty and Muller, 2021](#) D. Litty, V. Muller

Butyrate production in the acetogen Eubacterium limosum is dependent on the carbon and energy source

Microbial Biotechnology, 14 (6) (2021), pp. 2686-2692, [10.1111/1751-7915.13779](#) ↗

[View in Scopus](#) ↗

[Google Scholar](#) ↗

[Liu, Gao, Yu, Wu and Li, 2023](#) W. Liu, B. Gao, F. Yu, X. Wu, L. Li

Editorial: Edible mushrooms and the gut microbiota

Frontiers in Nutrition, 10 (2023), Article 1349429, [10.3389/fnut.2023.1349429](#) ↗

[View in Scopus](#) ↗

[Google Scholar](#) ↗

[Liu, Gibson and Walton, 2016](#) Y. Liu, G.R. Gibson, G.E. Walton

An in vitro approach to study effects of prebiotics and probiotics on the faecal microbiota and selected immune parameters relevant to the elderly

PLoS One, 11 (9) (2016), Article e0162604, [10.1371/journal.pone.0162604](#) ↗

[View in Scopus](#) ↗

[Google Scholar](#) ↗

[Martin-Gallausiaux, Marinelli, Blottiere, Larraufie and Lapaque, 2021](#) C. Martin-Gallausiaux, L. Marinelli, H.M. Blottiere, P. Larraufie, N. Lapaque

SCFA: Mechanisms and functional importance in the gut

The Proceedings of the Nutrition Society, 80 (1) (2021), pp. 37-49, [10.1017/S0029665120006916](#) ↗

[View in Scopus](#) ↗

[Google Scholar](#) ↗

[Mazhar, Zhu and Qin, 2023](#) M. Mazhar, Y. Zhu, L. Qin

The interplay of dietary fibers and intestinal microbiota affects type 2 diabetes by

generating short-chain fatty acids

Foods, 12 (5) (2023), p. 1023, [10.3390/foods12051023](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[McCauley et al., 2020](#) H.A. McCauley, A.L. Matthis, J.R. Enriquez, J.T. Nichol, J.G. Sanchez, W.J. Stone, ..., J.M. Wells

Enteroendocrine cells couple nutrient sensing to nutrient absorption by regulating ion transport

Nature Communications, 11 (1) (2020), p. 4791, [10.1038/s41467-020-18536-z](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[McMurdie and Holmes, 2013](#) P.J. McMurdie, S. Holmes

Phyloseq: An R package for reproducible interactive analysis and graphics of microbiome census data

PLoS One, 8 (4) (2013), Article e61217, [10.1371/journal.pone.0061217](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Mitsou et al., 2020](#) E.K. Mitsou, G. Saxami, E. Stamoulou, E. Kerezoudi, E. Terzi, G. Koutrotsios, ..., A. Kyriacou

Effects of rich in beta-glucans edible mushrooms on aging gut microbiota characteristics: An in vitro study

Molecules, 25 (12) (2020), p. 2806, [10.3390/molecules25122806](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Molly, Vande Woestyne and Verstraete, 1993](#) K. Molly, M. Vande Woestyne, W. Verstraete

Development of a 5-step multi-chamber reactor as a simulation of the human intestinal microbial ecosystem

Applied Microbiology and Biotechnology, 39 (2) (1993), pp. 254-258, [10.1007/BF00228615](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Moumita and Das, 2022](#) S. Moumita, B. Das

Assessment of the prebiotic potential and bioactive components of common edible mushrooms in India and formulation of synbiotic microcapsules

LWT, 156 (2022), Article 113050, [10.1016/j.lwt.2021.113050](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Muller et al., 2019](#) T.D. Muller, B. Finan, S.R. Bloom, D. D'Alessio, D.J. Drucker, P.R. Flatt, ..., M.H. Tschop

Glucagon-like peptide 1 (GLP-1)

Molecular Metabolism, 30 (2019), pp. 72-130, [10.1016/j.molmet.2019.09.010](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Mutuyemungu, Singh, Liu and Rose, 2023](#) E. Mutuyemungu, M. Singh, S. Liu, D.J. Rose

Intestinal gas production by the gut microbiota: A review

Journal of Functional Foods, 100 (2023), Article 105367, [10.1016/j.jff.2022.105367](https://doi.org/10.1016/j.jff.2022.105367) ↗

[View PDF](#)[View article](#)[View in Scopus](#) ↗[Google Scholar](#) ↗

[Natividad et al., 2018](#) J.M. Natividad, B. Lamas, H.P. Pham, M.L. Michel, D. Rainteau, C. Bridonneau, ..., H. Sokol

Bilophila wadsworthia aggravates high fat diet induced metabolic dysfunctions in mice

Nature Communications, 9 (1) (2018), p. 2802, [10.1038/s41467-018-05249-7](https://doi.org/10.1038/s41467-018-05249-7) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Plekhova et al., 2021](#) V. Plekhova, L. Van Meulebroek, M. De Graeve, A. Perdones-Montero, M. De Spiegeleer, E. De Paepe, ..., L. Vanhaecke

Rapid ex vivo molecular fingerprinting of biofluids using laser-assisted rapid evaporative ionization mass spectrometry

Nature Protocols, 16 (9) (2021), pp. 4327-4354, [10.1038/s41596-021-00580-8](https://doi.org/10.1038/s41596-021-00580-8) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Pompei et al., 2007](#) A. Pompei, L. Cordisco, A. Amaretti, S. Zanoni, D. Matteuzzi, M. Rossi

Folate production by bifidobacteria as a potential probiotic property

Applied and Environmental Microbiology, 73 (2007), pp. 179-185, [10.1128/AEM.01763-06](https://doi.org/10.1128/AEM.01763-06) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Possemiers, Verthe, Uyttendaele and Verstraete, 2004](#) S. Possemiers, K. Verthe, S. Uyttendaele, W. Verstraete
PCR-DGGE-based quantification of stability of the microbial community in a simulator of the human intestinal microbial ecosystem

FEMS Microbiology Ecology, 49 (3) (2004), pp. 495-507, [10.1016/j.femsec.2004.05.002](https://doi.org/10.1016/j.femsec.2004.05.002) ↗

[View PDF](#)[View article](#)[View in Scopus](#) ↗[Google Scholar](#) ↗

[Rios-Covian, Gueimonde, Duncan, Flint and de los Reyes-Gavilan, C. G., 2015](#) D. Rios-Covian, M. Gueimonde, S.H. Duncan, H.J. Flint, de los Reyes-Gavilan, C. G.

Enhanced butyrate formation by cross-feeding between Faecalibacterium prausnitzii and Bifidobacterium adolescentis

FEMS Microbiology Letters, 362 (21) (2015), Article fnv176, [10.1093/femsle/fnv176](https://doi.org/10.1093/femsle/fnv176) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Roberfroid, 1997](#) M.B. Roberfroid

Health benefits of non-digestible oligosaccharides

Advances in Experimental Medicine and Biology, 427 (1997), pp. 211-219, [10.1007/978-1-4615-5967-2_22](https://doi.org/10.1007/978-1-4615-5967-2_22) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Segata et al., 2011](#) N. Segata, J. Izard, L. Waldron, D. Gevers, L. Miropolsky, W.S. Garrett, C. Huttenhower
Metagenomic biomarker discovery and explanation

Genome Biology, 12 (6) (2011), p. R60, [10.1186/gb-2011-12-6-r60](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Singh, Saini, Kumar, Chawla and Kaushik, 2025](#) A. Singh, R.K. Saini, A. Kumar, P. Chawla, R. Kaushik
Mushrooms as nutritional powerhouses: A review of their bioactive compounds, health benefits, and value-added products

Foods, 14 (5) (2025), p. 741, [10.3390/foods14050741](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Song et al., 2022](#) L. Song, Q. Sun, H. Zheng, Y. Zhang, Y. Wang, S. Liu, L. Duan
Roseburia hominis alleviates neuroinflammation via short-chain fatty acids through histone deacetylase inhibition

Molecular Nutrition & Food Research, 66 (18) (2022), Article e2200164, [10.1002/mnfr.202200164](#) ↗

[Google Scholar](#) ↗

[Terry and Margolis, 2017](#) N. Terry, K.G. Margolis
Serotonergic mechanisms regulating the GI tract: Experimental evidence and therapeutic relevance

Handbook of Experimental Pharmacology, 239 (2017), pp. 319-342, [10.1007/164_2016_103](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Umetrics, 2012](#) MKS Umetrics. (2012). User guide to SIMCA. Retrieved 23 May 2025 from
<https://dl.icdst.org/pdfs/files3/18dae91414bba1b54de86520a34c0959.pdf> ↗.

[Google Scholar](#) ↗

[Van de Wiele, Boon, Possemiers, Jacobs and Verstraete, 2004](#) T. Van de Wiele, N. Boon, S. Possemiers, H. Jacobs, W. Verstraete
Prebiotic effects of chicory inulin in the simulator of the human intestinal microbial ecosystem

FEMS Microbiology Ecology, 51 (1) (2004), pp. 143-153, [10.1016/j.femsec.2004.07.014](#) ↗

[Google Scholar](#) ↗

[Vandeputte et al., 2017](#) D. Vandeputte, G. Falony, S. Vieira-Silva, J. Wang, M. Sailer, S. Theis, K. Verbeke, J. Raes
Prebiotic inulin-type fructans induce specific changes in the human gut microbiota

Gut, 66 (11) (2017), pp. 1968-1974, [10.1136/gutjnl-2016-313271](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Verhoeven, Keller, Verbruggen, Abboud and Venema, 2021](#) J. Verhoeven, D. Keller, S. Verbruggen, K.Y. Abboud, K. Venema
A blend of 3 mushrooms dose-dependently increases butyrate production by the gut microbiota

Benef Microbes, 12 (6) (2021), pp. 601-612, [10.3920/BM2021.0015](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Waqas et al., 2023](#) M. Waqas, S.A. Halim, A. Ullah, A.A.M. Ali, A. Khalid, A.N. Abdalla, ..., A. Al-Harrasi
Multi-fold computational analysis to discover novel putative inhibitors of
isethionate sulfite-lyase (isla) from *Bilophila wadsworthia*: Combating colorectal
cancer and inflammatory bowel diseases

Cancers (Basel), 15 (3) (2023), p. 901, [10.3390/cancers15030901](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Yao et al., 2021](#) C. Yao, Z. Wang, H. Jiang, R. Yan, Q. Huang, Y. Wang, ..., L. Lv
Ganoderma lucidum promotes sleep through a gut microbiota-dependent and
serotonin-involved pathway in mice

Scientific Reports, 11 (1) (2021), p. 13660, [10.1038/s41598-021-92913-6](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Yu et al., 2023](#) C. Yu, Q. Dong, M. Chen, R. Zhao, L. Zha, Y. Zhao, M. Zhang, B. Zhang, A. Ma
The effect of mushroom dietary fiber on the gut microbiota and related health
benefits: A review

J Fungi (Basel), 9 (10) (2023), p. 1028, [10.3390/jof9101028](#) ↗

[View in Scopus](#) ↗ [Google Scholar](#) ↗

[Zhang et al., 2025](#) L. Zhang, H. Wang, L. Zhao, J. Zhang, W. Sun, J. Chu, ..., D. Xu
Unraveling the interplay between mesenchymal stem cells, gut microbiota, and
systemic sclerosis: Therapeutic implications

Microbiol Spectr, e0157624 (2025), [10.1128/spectrum.01576-24](#) ↗

[Google Scholar](#) ↗

[Zhang et al., 2025](#) W. Zhang, Y. Zong, X. Huang, K. Liu, Z. Luo, J. Shan, L. Di
Cordyceps militaris alleviates COPD by regulating amino acid metabolism, gut
microbiota and short chain fatty acids

Journal of Ethnopharmacology, 346 (2025), Article 119701, [10.1016/j.jep.2025.119701](#) ↗



[View PDF](#) [View article](#) [View in Scopus](#) ↗ [Google Scholar](#) ↗

[Zhao et al., 2023](#) J. Zhao, Y. Hu, C. Qian, M. Hussain, S. Liu, A. Zhang, R. He, P. Sun
The interaction between mushroom polysaccharides and gut microbiota and their
effect on human health: A review

Biology (Basel), 12 (1) (2023), p. 122, [10.3390/biology12010122](#) ↗

[Google Scholar](#) ↗

[Zheng et al., 2022](#) H. Zheng, H. Cao, D. Zhang, J. Huang, J. Li, S. Wang, ..., X. Shi
Cordyceps militaris modulates intestinal barrier function and gut microbiota in a

pig model

Frontiers in Microbiology, 13 (2022), Article 810230, [10.3389/fmicb.2022.810230](https://doi.org/10.3389/fmicb.2022.810230) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

[Zhuang, Dong, Zhang and Feng, 2023](#) H. Zhuang, H. Dong, X. Zhang, T. Feng

Antioxidant activities and prebiotic activities of water-soluble, alkali-soluble polysaccharides extracted from the fruiting bodies of the fungus *Hericium erinaceus*

Polymers (Basel), 15 (20) (2023), p. 4165, [10.3390/polym15204165](https://doi.org/10.3390/polym15204165) ↗

[View in Scopus ↗](#) [Google Scholar ↗](#)

Cited by (0)

© 2025 The Authors. Published by Elsevier Ltd.

All content on this site: Copyright © 2026 Elsevier B.V., its licensors, and contributors. All rights are reserved, including those for text and data mining, AI training, and similar technologies. For all open access content, the relevant licensing terms apply.